

The Maximal Cardinality of Quantum Latin Cubes: A Complete Solution

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ABSTRACT. A quantum Latin cube (QLC) of order v is a $v \times v \times v$ array of unit vectors in $\mathbb{C}^v$ such that every line is an orthonormal basis. Its cardinality is the number of projectively distinct entries, at most v^3 . We prove that this maximum is attainable if and only if $v = 1$ or $v \geq 4$, closing the previously open prime orders and orders pq with p, q prime and $(p, q) \neq (2, 2)$. For every prime $p \geq 5$, an explicit tri-clock construction reduces maximality to linear independence of four functions over $\mathbb{F}_p$. Composite orders are handled by twisted layers over flat quantum Latin squares. We also characterise all layered QLCs in terms of cyclic simple spectrum and flatness of the base square. Consequently, layered cardinalities are divisible by v , while the prime tri-clock family has exact spectrum $\{p, p^2, p^3\}$.

1. INTRODUCTION

A *quantum Latin square* of order v , denoted $\text{QLS}(v)$, is a $v \times v$ array whose entries are unit column vectors of the Hilbert space $\mathcal{H}_v = \mathbb{C}^v$ such that the v entries of every row and of every column form an orthonormal basis. Quantum Latin squares were introduced by Musto and Vicary [2] and have since found applications to unitary error bases, mutually unbiased bases, quantum Sudoku and absolutely maximally entangled states [3, 4, 5, 1]. Two unit vectors $|u\rangle, |v\rangle$ are identified when $|u\rangle = e^{i\theta}|v\rangle$ for some real θ . Following Paczos, Wierzbński, Rajchel-Mieldzioć, Burchardt and Życzkowski [5], the *cardinality* c of a $\text{QLS}(v)$ is the number of distinct vectors appearing in the array; clearly $v \leq c \leq v^2$. After intensive recent activity [5, 6, 7, 8, 9, 10] the existence problem for squares of maximal cardinality is completely solved: a $\text{QLS}(v)$ with cardinality v^2 exists if and only if $v = 1$ or $v \geq 4$ [5, 6, 8].

Quantum Latin *cubes* were introduced by Goyeneche, Raissi, Di Martino and Życzkowski [1] in connection with absolutely maximally entangled states. We use the definition of [7, Def. 1.2].

Definition 1.1. A *quantum Latin cube* of order v , denoted $\text{QLC}(v)$, is a $v \times v \times v$ array $C = (|c_{i,j,k}\rangle)_{i,j,k \in \mathbb{Z}_v}$ of unit vectors of $\mathbb{C}^v$ such that each *line* of C — the v entries obtained by fixing two of the three indices — forms an orthonormal basis of $\mathbb{C}^v$. The *cardinality* $c(C)$ is the number of distinct entries of C up to global phase; clearly $v \leq c(C) \leq v^3$.

Replacing each symbol of a classical Latin cube by the corresponding computational basis vector yields a $\text{QLC}(v)$ of minimal cardinality v . At the opposite extreme, Zhang, Zhang, Lv and Cao proved the following.

Theorem 1.2 ([7, Thm. 1.3]). *There exists a $\text{QLC}(v)$ with maximal cardinality v^3 for all integers $v \geq 4$, except when v is a prime or $v = pq$ with p, q primes and $(p, q) \neq (2, 2)$.*

Thus the existence of quantum Latin cubes of maximal cardinality remained open for all primes $v \geq 5$ and for all orders pq with p, q prime and $(p, q) \neq (2, 2)$, the smallest open orders being $v = 5, 6, 7, 9, 10$. In this paper we resolve all remaining orders and obtain a complete answer, which exactly parallels the dichotomy known for squares.

This is a cube endpoint problem, not another square-spectrum theorem. The recent square results cited above settle the maximal endpoint for $\text{QLS}(v)$ and, for several infinite families, long or complete intervals of square cardinalities. They do not construct three-dimensional arrays with v^3 projectively distinct entries in the prime and pq cases left open by Theorem 1.2. Conversely, our main theorem determines only the maximal endpoint for cubes; it does not claim a dense spectrum for $\text{QLC}(v)$, nor does it classify orthogonal families of quantum Latin squares or cubes.

Theorem 1.3. *A quantum Latin cube of order v with maximal cardinality v^3 exists if and only if $v = 1$ or $v \geq 4$.*

The proof has three components, each with a different role: a rigidity theorem for layered cubes, an explicit construction for prime orders, and a twisted-product construction for composite orders.

Prime orders: the tri-clock construction. For a prime p and permutations π, g of $\mathbb{Z}_p$, define a cube whose entry at position (r, j, k) is the unit vector

$$C_{r,j,k} = \frac{1}{\sqrt{p}} \left(\omega^{rs+j\pi(s)+kg(s)} \right)_{s \in \mathbb{Z}_p}, \quad \omega = e^{2\pi i/p}.$$

Orthonormality along the three axes is immediate from the bijectivity of $s \mapsto s, \pi$ and g , and we show (Theorem 5.1) that C has maximal cardinality p^3 if and only if the four functions $1, s, \pi(s), g(s)$ are linearly independent over the field $\mathbb{F}_p$. More generally, its cardinality is p^r , where r is the rank of $s, \pi(s), g(s)$ modulo the constant functions; this determines the exact tri-clock spectrum $\{p, p^2, p^3\}$ for every $p \geq 5$. A single explicit pair, with π the transposition of $p-2$ and $p-1$ and g the 3-cycle $0 \mapsto 1 \mapsto 2 \mapsto 0$, satisfies this condition for *every* prime $p \geq 5$ (Lemma 5.2), so the construction is completely explicit and its verification is elementary.

Composite orders: twisted layers. For composite $v = mn \geq 6$ we start from the product construction of Zhang and Cao [6, 7], which assembles a $\text{QLS}(mn)$ of maximal cardinality out of row-quantum Latin rectangles, and we choose the rectangles inside *flat* (Fourier-type) bases, so that every entry of the resulting square is unbiased to the computational basis. Stacking the v layers $V^k Q$, $k \in \mathbb{Z}_v$, for a suitable diagonal unitary V then produces a $\text{QLC}(v)$, and a combinatorial choice of the eigenvalue assignment of V forces all v^3 entries to be distinct whenever one 2×2 double difference is a unit in $\mathbb{Z}_v$ (Theorem 4.3). Thus the composite-order construction is a family, not a single example: for each factorisation $v = mn$ with $m, n \geq 2$ and $v \geq 6$ there are at least $v\varphi(v)(v-4)!$ admissible eigenvalue assignments (Corollary 4.4; inequivalence is not asserted). The layered architecture is also forced in the following precise sense: we prove a rigidity theorem (Theorem 3.3) showing that an array of the form $(V^k q_{ij})_{i,j,k}$ is a quantum Latin cube *precisely when* the spectrum of V is a rotated full set of v -th roots of unity and the layer $Q = (q_{ij})$ is a quantum Latin square all of whose entries are unbiased to the eigenbasis of V . In particular the tri-clock cubes are exactly of this form: their first layers are, up to relabelling, the maximal-cardinality squares constructed by Zang, Zheng, Tian and Shan [8, Thm. 3.1] from phased Fourier matrices (Corollary 5.7 and Remark 5.8). Those squares are unbiased to a fixed basis, and the rigidity theorem identifies this flatness as precisely the property that makes a maximal-cardinality square liftable to a cube. This gives a structural explanation for why this family, rather than any other known family of maximal-cardinality squares, is the right base for cube constructions.

The same framework gives two obstructions that delimit the reach of these methods. Every layered cube has cardinality divisible by its order (Corollary 3.6), and a prime-order tri-clock cube has cardinality p^r , where r is the rank of three functions modulo the constant functions (Theorem 5.1). All three ranks occur, so the tri-clock family has spectrum exactly $\{p, p^2, p^3\}$

for $p \geq 5$ (Corollary 5.4). Thus neither construction can by itself account for an arbitrary middle-spectrum value.

Finally, the same rigidity viewpoint explains why order four is genuinely special: we show by a parity argument over $\mathbb{F}_2$ that no pair of permutations of $\mathbb{Z}_4$ can satisfy the tri-clock independence condition (Proposition 6.3), so the cube of [7, Ex. 4.1] cannot be obtained by the tri-clock recipe; and we prove that quantum Latin cubes of orders 2 and 3 are themselves necessarily classical, so that their cardinality spectra are $\{2\}$ and $\{3\}$ (Theorem 6.2). The key step for order 3 is the elementary fact that a 3×3 unitary matrix with zero diagonal must be monomial (Lemma 6.1).

As a supplementary application of the same projective-cardinality viewpoint, we study quantum Latin squares of order five. Combining exact constructions with finite support–orthogonality certificates gives $\text{Spec}_{\text{QLS}}(5) \cap [5, 12] = \{5, 7, 12\}$ (Theorem 6.7). Besides sharpening the small-order picture, this calculation supplies a reproducible obstruction calculus for attacking the still-open middle part of the square spectrum. This square-spectrum result is logically independent of the proof of Theorem 1.3.

The small-order square spectrum is changing quickly. The dense-spectrum theorems of Zhang, Wang and Ji [9] and Zhang and Ji [10] establish all possible cardinalities for the infinite families of orders $4m$ and $6m$ with $m \geq 2$, respectively, while Xu gives explicit QLS(6) of cardinalities 13, 15 and 17 [11]. Thus the frontier for squares is no longer merely the two endpoints; it is a mixture of dense high-divisibility results, isolated small-order constructions, and finite obstruction problems. This is the context in which the certified order-five interval below should be read.

We distinguish throughout three levels of evidence. The maximal-cardinality cube theorem is proved by explicit formulae and finite algebraic checks. The order-five low-spectrum theorem is computer-assisted: its nonexistence claims depend on a finite exhaustive enumeration of labelled support–orthogonality types, after which every terminal type is eliminated by one of the elementary linear-algebra configurations in Lemma 6.6. Numerical searches mentioned in remarks and problems are used only to guide conjectures; they are not evidence for any theorem.

The paper is organised as follows. Section 2 fixes notation and recalls the results we use. Section 3 develops the twisted-layer framework and the rigidity theorem. Section 4 treats composite orders and Section 5 prime orders; Theorem 1.3 is assembled at the end of Section 5. Section 6 contains the order-four obstruction, the small orders, the certified low spectrum of QLS(5), and open problems on cardinality spectra.

2. PRELIMINARIES

Throughout, $v \geq 2$ is an integer, $\omega_v = e^{2\pi i/v}$, and $\{|0\rangle, \dots, |v-1\rangle\}$ is the computational basis of $\mathbb{C}^v$. Indices of arrays are taken in $\mathbb{Z}_v$ and index arithmetic is modular. For a unit vector $|q\rangle$ and an orthonormal basis $B = \{|b_0\rangle, \dots, |b_{v-1}\rangle\}$ we say $|q\rangle$ is *flat* (or *unbiased*) with respect to B if $|\langle b_t, q \rangle|^2 = 1/v$ for all t . The *Fourier matrix* is $F_v = \frac{1}{\sqrt{v}}(\omega_v^{st})_{s,t \in \mathbb{Z}_v}$; its columns form an orthonormal basis of flat vectors.

If Q is a QLS(v) (or a QLC(v)) and $U \in \text{U}(v)$, the entrywise image UQ is again a QLS(v) (resp. QLC(v)) with the same cardinality, since unitaries preserve inner products and projective distinctness.

We will use the following known facts.

Lemma 2.1 ([5]). *Every QLS(2) has cardinality 2 and every QLS(3) has cardinality 3; equivalently, quantum Latin squares of orders 2 and 3 are classical up to a global unitary.*

Lemma 2.2 ([6, Lem. 2.6]). *Let $|a\rangle, |b\rangle \in \mathbb{C}^n$ and $|c\rangle, |d\rangle \in \mathbb{C}^m$ be unit vectors. Then $|a\rangle \otimes |c\rangle$ and $|b\rangle \otimes |d\rangle$ are equal up to phase if and only if $|a\rangle, |b\rangle$ are equal up to phase and $|c\rangle, |d\rangle$ are equal up to phase.*

An $m \times n$ row-quantum Latin rectangle in $\mathbb{C}^n$ is an $m \times n$ array of unit vectors of $\mathbb{C}^n$ each of whose rows is an orthonormal basis of $\mathbb{C}^n$; its cardinality is the number of distinct entries.

Lemma 2.3 ([6, Thm. 3.1]; see also [7, Lem. 2.2]). *Let $U = (|u_{i,k}\rangle)$ be an $m \times n$ row-quantum Latin rectangle in $\mathbb{C}^n$ with cardinality c_1 and $V = (|v_{j,l}\rangle)$ an $n \times m$ row-quantum Latin rectangle in $\mathbb{C}^m$ with cardinality c_2 . Let W be the array of order mn partitioned into an $m \times n$ grid of $n \times m$ blocks, whose entry in position (k, l) of block (i, j) is*

$$|w_{i,j,k,l}\rangle = |u_{i,j+k}\rangle \otimes |v_{j,i+l}\rangle, \quad j+k \bmod n, \quad i+l \bmod m.$$

Then W is a QLS(mn) with cardinality $c_1 c_2$.

We also record for context that quantum Latin squares of maximal cardinality exist in every order $v \geq 4$ [8]. Our constructions below are self-contained and do not rely on this; the layer square of the tri-clock cube does, however, turn out to coincide with the construction of [8, Thm. 3.1] — see Remark 5.8.

3. TWISTED LAYERS AND A RIGIDITY THEOREM

The constructions of this paper produce a cube by stacking the iterates of a fixed unitary applied to a single square. In this section we determine exactly when this is possible.

Definition 3.1. Let $V \in \mathcal{U}(v)$ and let $Q = (|q_{i,j}\rangle)_{i,j \in \mathbb{Z}_v}$ be an array of unit vectors of $\mathbb{C}^v$. The V -layered cube over Q is the array $C = C(V, Q)$ with entries

$$|c_{i,j,k}\rangle = V^k |q_{i,j}\rangle, \quad i, j, k \in \mathbb{Z}_v.$$

The lines of C in the first two directions are the images under the unitary V^k of the rows and columns of Q , so they are orthonormal bases if and only if Q is a QLS(v). The lines in the third direction — the *orbit lines* $\{V^k |q\rangle\}_{k \in \mathbb{Z}_v}$ — are governed by the following two lemmas.

Lemma 3.2. *For a unit vector $|q\rangle \in \mathbb{C}^v$ and $V \in \mathcal{U}(v)$, the orbit $\{V^k |q\rangle : k = 0, 1, \dots, v-1\}$ is an orthonormal basis of $\mathbb{C}^v$ if and only if $\langle q, V^k q \rangle = 0$ for $k = 1, \dots, v-1$.*

Proof. $\langle V^a q, V^b q \rangle = \langle q, V^{b-a} q \rangle$ and, for $0 \leq a < b \leq v-1$, the exponent $b-a$ ranges over $1, \dots, v-1$. $\square$

Theorem 3.3 (Rigidity). *Let $V \in \mathcal{U}(v)$ and suppose some orthonormal basis $\{|q_1\rangle, \dots, |q_v\rangle\}$ of $\mathbb{C}^v$ has the property that every orbit $\{V^k |q_j\rangle\}_{k \in \mathbb{Z}_v}$ is an orthonormal basis. Then:*

- (i) V has v distinct eigenvalues, and they form a rotated full set of v -th roots of unity: there is a unimodular μ and an eigenbasis $B = \{|b_t\rangle\}_{t \in \mathbb{Z}_v}$ with $V|b_t\rangle = \mu \omega_v^t |b_t\rangle$;
- (ii) every unit vector $|q\rangle$ whose orbit $\{V^k |q\rangle\}_k$ is an orthonormal basis is flat with respect to B .

Proof. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of V with spectral projections $P_1, \dots, P_r$, and for a unit vector $|q\rangle$ put $w_u(q) = \|P_u |q\rangle\|^2$, so that $w_u(q) \geq 0$, $\sum_u w_u(q) = 1$ and

$$(1) \quad \langle q, V^k q \rangle = \sum_{u=1}^r w_u(q) \lambda_u^k, \quad k \in \mathbb{Z}.$$

Fix j and abbreviate $w_u = w_u(q_j)$. By Lemma 3.2, $\sum_u w_u \lambda_u^k = 0$ for $k = 1, \dots, v-1$.

Suppose $r \leq v-1$. Then the equations with $k = 1, \dots, r$ are available, and they say that the vector $(w_u \lambda_u)_u$ is annihilated by the Vandermonde matrix $(\lambda_u^{k-1})_{k=1, \dots, r; u=1, \dots, r}$, which is invertible because the λ_u are distinct. Hence $w_u \lambda_u = 0$, so $w_u = 0$ for all u , contradicting $\sum_u w_u = 1$. Therefore $r = v$: all eigenvalues of V are simple, and the eigenbasis $B = \{|b_t\rangle\}$ is determined up to phases; write $w_t(q) = |\langle b_t, q \rangle|^2$.

Now the full system $\sum_t w_t \lambda_t^k = \delta_{k,0}$ ($k = 0, 1, \dots, v-1$) is a square Vandermonde system, so it has a *unique* solution $w^* = (w_t^*)_t$, and $w(q) = w^*$ for every unit vector $|q\rangle$ whose orbit is an orthonormal basis — in particular for each $|q_j\rangle$. Since $\{|q_j\rangle\}_j$ is an orthonormal basis, Parseval gives, for each t ,

$$v w_t^* = \sum_{j=1}^v |\langle b_t, q_j \rangle|^2 = \|b_t\|^2 = 1,$$

so $w_t^* = 1/v$: every such $|q\rangle$ is flat with respect to B , proving (ii). Finally, (1) with $w = w^*$ yields $\sum_t \lambda_t^k = 0$ for $k = 1, \dots, v-1$, i.e. the first $v-1$ power sums of the λ_t vanish; by Newton's identities all elementary symmetric functions $e_1, \dots, e_{v-1}$ vanish as well, so the characteristic polynomial of V is $z^v - (-1)^{v+1} e_v$ and the λ_t are the v -th roots of a single unimodular number. This is (i). $\square$

Theorem 3.4 (Characterisation of layered cubes). *$C(V, Q)$ is a QLC(v) if and only if*

- (a) *Q is a QLS(v);*
- (b) *the spectrum of V is a rotated full set of v -th roots of unity, $\{\mu \omega_v^t : t \in \mathbb{Z}_v\}$, with all eigenvalues simple;*
- (c) *every entry of Q is flat with respect to the eigenbasis of V .*

Proof. ($\Leftarrow$) Lines in the first two directions are unitary images of rows and columns of Q . For an orbit line, (b) and (c) give, for $k \neq 0$,

$$\langle q, V^k q \rangle = \sum_t |\langle b_t, q \rangle|^2 (\mu \omega_v^t)^k = \frac{\mu^k}{v} \sum_{t \in \mathbb{Z}_v} \omega_v^{tk} = 0,$$

and Lemma 3.2 applies. ($\Rightarrow$) The layer $k = 0$ of a QLC is a QLS, giving (a). The first row of Q is an orthonormal basis each of whose orbits is a line of C , hence an orthonormal basis; Theorem 3.3 gives (b), and (ii) of that theorem applies to every entry of Q (each entry's orbit is a line of C), giving (c). $\square$

Lemma 3.5 (Cardinality of layered cubes). *In the situation of Theorem 3.4, write $|q\rangle \sim_V |q'\rangle$ if $|q'\rangle$ is proportional to $V^k |q\rangle$ for some k . Then each projective V -orbit $\{[V^k |q]\} : k \in \mathbb{Z}_v\}$ has exactly v elements, and*

$$c(C(V, Q)) = v \cdot \#(\{\text{entries of } Q\} / \sim_V).$$

In particular $c(C) = v^3$ if and only if $c(Q) = v^2$ and no two distinct entries of Q are $\sim_V$ -equivalent.

Proof. If $V^k |q\rangle = e^{i\theta} |q\rangle$ with $0 < k < v$, then comparing coordinates in the eigenbasis and using that a flat vector has no zero coordinate, $\mu^k \omega_v^{tk} = e^{i\theta}$ for all t , whence $\omega_v^{(t-t')k} = 1$ for all t, t' , i.e. $v \mid k$, a contradiction. So all orbits are free, and two orbits are either disjoint or equal as projective sets. The entries of C are precisely the union of the orbits of the entries of Q , and the count follows. $\square$

Corollary 3.6 (Layered divisibility obstruction). *The cardinality of every layered QLC(v) is divisible by v . Hence a QLC(v) whose cardinality is not divisible by v admits no representation $C(V, Q)$, even after permuting the three coordinate directions.*

Proof. The divisibility follows from Lemma 3.5. Permuting the coordinate directions preserves cardinality, so the same obstruction applies to a layered representation in any direction. $\square$

4. COMPOSITE ORDERS

4.1. Flat Fourier bases.

Lemma 4.1. *For $n \geq 2$ and $i \in \mathbb{Z}_{\geq 0}$ define unit vectors $|x_a^{(i)}\rangle \in \mathbb{C}^n$, $a \in \mathbb{Z}_n$, by*

$$(x_a^{(i)})_s = \frac{1}{\sqrt{n}} e^{i\sqrt{2} \cdot i \delta_{s,0}} \omega_n^{sa}, \quad s \in \mathbb{Z}_n,$$

that is, $B_i := \{|x_a^{(i)}\rangle : a \in \mathbb{Z}_n\}$ consists of the columns of $D_i F_n$ where $D_i = \text{diag}(e^{i\sqrt{2}i}, 1, \dots, 1)$. Then each B_i is an orthonormal basis all of whose vectors are flat with respect to the computational basis, and for $i \neq i'$ the bases B_i and $B_{i'}$ have no projective vector in common.

Proof. $D_i F_n$ is unitary with all entries of modulus $n^{-1/2}$, giving the first claim. Suppose $|x_a^{(i)}\rangle = e^{i\theta} |x_{a'}^{(i')}\rangle$. Comparing coordinates,

$$(2) \quad \sqrt{2}(i - i')\delta_{s,0} + \frac{2\pi s}{n}(a - a') \equiv \theta \pmod{2\pi} \quad \text{for all } s \in \mathbb{Z}_n.$$

If $n \geq 3$, the cases $s = 1, 2$ give $\frac{2\pi}{n}(a - a') \equiv 0$, so $a = a'$ and then $\theta \equiv 0$, and $s = 0$ gives $\sqrt{2}(i - i') \in 2\pi\mathbb{Z}$; since π is transcendental this forces $i = i'$. If $n = 2$, the case $s = 1$ gives $\theta \equiv \pi(a - a')$, so $\theta \in \pi\mathbb{Z}$, and $s = 0$ gives $\sqrt{2}(i - i') \in \pi\mathbb{Z}$, again forcing $i = i'$ and then $a = a'$. $\square$

4.2. A flat square of maximal cardinality. Let $v = mn$ with $m, n \geq 2$. Apply Lemma 4.1 twice: in dimension n it yields bases $B_0, \dots, B_{m-1}$ of $\mathbb{C}^n$, giving the $m \times n$ row-quantum Latin rectangle $U = (|u_{i,a}\rangle)$ with $|u_{i,a}\rangle = |x_a^{(i)}\rangle$; in dimension m it yields bases $B'_0, \dots, B'_{n-1}$ of $\mathbb{C}^m$, giving the $n \times m$ row-quantum Latin rectangle $V = (|v_{j,b}\rangle)$. All mn entries of U are pairwise projectively distinct, and likewise for V , so $c_1 = c_2 = mn$. Let W be the QLS(mn) of Lemma 2.3; then $c(W) = (mn)^2$ is maximal. Moreover:

- (1) every entry of W equals $|u_{i,a}\rangle \otimes |v_{j,b}\rangle$ for some $(i, a) \in \mathbb{Z}_m \times \mathbb{Z}_n$ and $(j, b) \in \mathbb{Z}_n \times \mathbb{Z}_m$, and conversely every such tensor occurs as an entry (in block (i, j) , at inner position $k = a - j, l = b - i$);
- (2) every entry of W is flat with respect to the computational product basis

$$\{|s\rangle \otimes |t\rangle : (s, t) \in \mathbb{Z}_n \times \mathbb{Z}_m\},$$

because each coordinate of $|u_{i,a}\rangle \otimes |v_{j,b}\rangle$ has modulus $n^{-1/2}m^{-1/2} = v^{-1/2}$.

4.3. The twist.

Lemma 4.2. *For $v = mn \geq 6$ with $m, n \geq 2$ there exists a bijection $g : \mathbb{Z}_n \times \mathbb{Z}_m \rightarrow \mathbb{Z}_v$ with a unit double difference, i.e.*

$$\Delta g := g(1, 1) - g(1, 0) - g(0, 1) + g(0, 0) \in \mathbb{Z}_v^\times.$$

In fact one may prescribe

$$g(0, 0) = 0, \quad g(1, 0) = 1, \quad g(0, 1) = 2, \quad g(1, 1) = 4.$$

Consequently, for each fixed factorisation $v = mn$, there are at least $v\varphi(v)(v-4)!$ such bijections, where φ is Euler's function.

Proof. The four cells exist since $m, n \geq 2$, the four prescribed values are distinct elements of $\mathbb{Z}_v$ since $v \geq 5$, and the remaining $v - 4$ values $\{3, 5, 6, \dots, v - 1\}$ may be assigned to the remaining cells arbitrarily. For every such assignment $\Delta g = 4 - 1 - 2 + 0 = 1$, a unit in $\mathbb{Z}_v$.

This gives $(v-4)!$ normalised bijections. If $a \in \mathbb{Z}_v^\times$ and $b \in \mathbb{Z}_v$, then $ag+b$ is again a bijection and

$$\Delta(ag+b) = a \Delta g \in \mathbb{Z}_v^\times.$$

The resulting $v\varphi(v)(v-4)!$ bijections are distinct: from $h = ag+b$ one recovers $b = h(0,0)$ and $a = h(1,0) - h(0,0)$, and then recovers g . $\square$

Theorem 4.3. *Let $v = mn \geq 6$ with $m, n \geq 2$, and let $g : \mathbb{Z}_n \times \mathbb{Z}_m \rightarrow \mathbb{Z}_v$ be any bijection with unit double difference $\Delta g \in \mathbb{Z}_v^\times$. Then the twisted-product construction below gives a $\text{QLC}(v)$ with maximal cardinality v^3 . In particular every composite integer $v \geq 6$ admits a $\text{QLC}(v)$ with maximal cardinality.*

Proof. Let W be the flat $\text{QLS}(v)$ of maximal cardinality constructed above, and let

$$V_g = \sum_{(s,t) \in \mathbb{Z}_n \times \mathbb{Z}_m} \omega_v^{g(s,t)} (|s\rangle \otimes |t\rangle)(\langle s| \otimes \langle t|).$$

The spectrum of V_g is the full set of v -th roots of unity with simple eigenvalues, and every entry of W is flat with respect to its eigenbasis (the computational product basis), so $C = C(V_g, W)$ is a $\text{QLC}(v)$ by Theorem 3.4.

It remains to verify, via Lemma 3.5, that distinct entries of W are $\sim_{V_g}$ -inequivalent. Suppose

$$V_g^\kappa (|u_{i,a}\rangle \otimes |v_{j,b}\rangle) = e^{i\theta} |u_{i',a'}\rangle \otimes |v_{j',b'}\rangle$$

for some $\kappa \in \mathbb{Z}_v$. All coordinates of both sides are nonzero, and comparing their phases at the coordinate (s,t) gives, for all $(s,t) \in \mathbb{Z}_n \times \mathbb{Z}_m$,

$$(3) \quad \frac{2\pi\kappa}{v} g(s,t) + A(s) + B(t) \equiv \theta \pmod{2\pi},$$

where $A(s) = \sqrt{2}(i-i')\delta_{s,0} + \frac{2\pi s}{n}(a-a')$ and $B(t) = \sqrt{2}(j-j')\delta_{t,0} + \frac{2\pi t}{m}(b-b')$ collect all terms depending on s alone, respectively on t alone. Evaluating (3) at the four cells $(0,0), (1,0), (0,1), (1,1)$ and taking the alternating sum annihilates θ , A and B entirely and leaves

$$\frac{2\pi\kappa}{v} [g(1,1) - g(1,0) - g(0,1) + g(0,0)] = \frac{2\pi\kappa \Delta g}{v} \equiv 0 \pmod{2\pi},$$

whence $v \mid \kappa \Delta g$. Since Δg is a unit modulo v , this gives $\kappa \equiv 0$. Then the two entries of W coincide projectively, and since $c(W) = v^2$, Lemma 2.2 and Lemma 4.1 force $(i,a,j,b) = (i',a',j',b')$. By Lemma 3.5, $c(C) = v \cdot v^2 = v^3$. The existence assertion follows by choosing g as in Lemma 4.2. $\square$

Corollary 4.4 (Abundance of composite twists). *For each factorisation $v = mn \geq 6$ with $m, n \geq 2$, the preceding construction has at least $v\varphi(v)(v-4)!$ admissible eigenvalue assignments g giving maximal-cardinality layered $\text{QLC}(v)$. This count is by assignments; no inequivalence of the resulting cubes is asserted.*

Proof. Use the assignments of Lemma 4.2 in Theorem 4.3. $\square$

Corollary 4.5. *For every composite integer $v \geq 4$ there exists a $\text{QLC}(v)$ with maximal cardinality.*

Proof. Combine Theorem 4.3 with the explicit $\text{QLC}(4)$ of maximal cardinality of [7, Ex. 4.1]. $\square$

Remark 4.6. The hypothesis $v \geq 6$ in Lemma 4.2 is sharp for this corner-twist method. If $g : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ is a bijection, then, modulo 2,

$$\Delta g \equiv g(1,1) + g(1,0) + g(0,1) + g(0,0) \equiv 0 + 1 + 0 + 1 \equiv 0,$$

so Δg is even and cannot be a unit in $\mathbb{Z}_4$. The tri-clock method has its own order-four obstruction, proved in Proposition 6.3.

5. PRIME ORDERS: THE TRI-CLOCK CONSTRUCTION

Throughout this section $p \geq 5$ is prime and $\omega = \omega_p$. We regard functions $\mathbb{Z}_p \rightarrow \mathbb{Z}_p$ as vectors of the p -dimensional space $\mathbb{F}_p^{\mathbb{Z}_p}$ over the field $\mathbb{F}_p$; the four functions of interest are the constant $\mathbf{1}$, the identity $\iota(s) = s$, and two permutations π, g of $\mathbb{Z}_p$.

Theorem 5.1. *Let p be prime and let π, g be permutations of $\mathbb{Z}_p$. Define the cube $C = (|c_{r,j,k}\rangle)_{r,j,k \in \mathbb{Z}_p}$ by*

$$|c_{r,j,k}\rangle = \frac{1}{\sqrt{p}} \left(\omega^{rs+j\pi(s)+kg(s)} \right)_{s \in \mathbb{Z}_p} \in \mathbb{C}^p.$$

Then C is a QLC(p). If

$$r_0 = \dim_{\mathbb{F}_p} \text{span}\{[\iota], [\pi], [g]\} \quad \text{in} \quad \mathbb{F}_p^{\mathbb{Z}_p} / \langle \mathbf{1} \rangle,$$

then

$$c(C) = p^{r_0}.$$

In particular $r_0 \in \{1, 2, 3\}$, so a tri-clock cube has cardinality p , p^2 or p^3 ; it has maximal cardinality if and only if $\{\mathbf{1}, \iota, \pi, g\}$ is linearly independent over $\mathbb{F}_p$.

Proof. Lines. For a line in the first direction (fix j, k , vary r),

$$\langle c_{r,j,k}, c_{r',j,k} \rangle = \frac{1}{p} \sum_{s \in \mathbb{Z}_p} \omega^{(r'-r)s} = \delta_{r,r'},$$

since $\sum_{s \in \mathbb{Z}_p} \omega^{cs} = 0$ for $c \not\equiv 0$. For the second direction the same computation applies after the substitution $u = \pi(s)$, which is a bijection of $\mathbb{Z}_p$: $\langle c_{r,j,k}, c_{r,j',k} \rangle = \frac{1}{p} \sum_u \omega^{(j'-j)u} = \delta_{j,j'}$; likewise for the third direction using g . Hence C is a QLC(p).

Cardinality. The projective class of the entry at (r, j, k) is determined by the coset of $r\iota + j\pi + kg$ modulo the constant functions. More explicitly, two entries coincide projectively, $|c_{r,j,k}\rangle = e^{i\theta} |c_{r',j',k'}\rangle$, if and only if the function

$$s \mapsto (r - r')s + (j - j')\pi(s) + (k - k')g(s) \pmod{p}$$

is constant on $\mathbb{Z}_p$, i.e. if and only if $(r - r')\iota + (j - j')\pi + (k - k')g \in \mathbb{F}_p \mathbf{1}$. Thus the projective classes of the entries are in bijection with the image of the linear map

$$\mathbb{F}_p^3 \longrightarrow \mathbb{F}_p^{\mathbb{Z}_p} / \langle \mathbf{1} \rangle, \quad (r, j, k) \mapsto [r\iota + j\pi + kg].$$

Its image has dimension r_0 , proving $c(C) = p^{r_0}$. Since ι is non-constant, $1 \leq r_0 \leq 3$. If $\{\mathbf{1}, \iota, \pi, g\}$ is linearly independent this forces $r = r', j = j', k = k'$, so all p^3 entries are distinct. Conversely, a nontrivial relation $a\mathbf{1} + b\iota + c\pi + dg = 0$ must have $(b, c, d) \neq (0, 0, 0)$ (else $a = 0$ as well), and then $bs + c\pi(s) + dg(s) \equiv -a$ is constant, so the entries at positions $(0, 0, 0)$ and (b, c, d) coincide projectively and $c(C) < p^3$. $\square$

Lemma 5.2. *For every prime $p \geq 5$, let π be the transposition exchanging $p-2$ and $p-1$ (and fixing all other points), and let g be the 3-cycle $g(0) = 1, g(1) = 2, g(2) = 0$ (fixing all other points). Then $\{\mathbf{1}, \iota, \pi, g\}$ is linearly independent over $\mathbb{F}_p$.*

Proof. Suppose $a + bs + c\pi(s) + dg(s) \equiv 0 \pmod{p}$ for all $s \in \mathbb{Z}_p$.

First let $p \geq 7$. The set $S = \{3, 4, \dots, p-3\}$ has $p-5 \geq 2$ elements, and for $s \in S$ we have $\pi(s) = g(s) = s$, so $a + (b+c+d)s \equiv 0$ at two or more distinct values of s ; hence $b+c+d \equiv 0$ and $a \equiv 0$. At $s = p-2$ and $s = p-1$ we have $g(s) = s$, so $(b+d)s + c\pi(s) \equiv 0$; subtracting the two instances gives $(b+d) - c \equiv 0$, i.e. $c \equiv b+d$. Combining with $b+c+d \equiv 0$ yields $2c \equiv 0$, so $c \equiv 0$ (as p is odd) and $d \equiv -b$. Finally $s = 0$ gives $d g(0) = d \equiv 0$, hence $b \equiv 0$. Thus $a = b = c = d = 0$.

For $p = 5$ we have $\pi = (0, 1, 2, 4, 3)$ and $g = (1, 2, 0, 3, 4)$ (in one-line notation), and the rows $s = 0, 1, 2, 3$ of the matrix $(\mathbf{1}, \iota, \pi, g)$ form

$$M = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 \\ 1 & 3 & 4 & 3 \end{pmatrix}, \quad \det M = \det \begin{pmatrix} 1 & 1 & 2 \\ 2 & 2 & 0 \\ 3 & 4 & 3 \end{pmatrix} - \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & 4 \end{pmatrix} = 4 - 1 = 3 \not\equiv 0 \pmod{5},$$

so the four columns are linearly independent over $\mathbb{F}_5$. $\square$

Corollary 5.3. *For every prime $p \geq 5$ there exists an explicit QLC(p) with maximal cardinality p^3 .*

Corollary 5.4 (Exact spectrum of the prime tri-clock family). *For every prime $p \geq 5$, the set of cardinalities attained by tri-clock QLC(p) is exactly*

$$\{p, p^2, p^3\}.$$

Proof. Theorem 5.1 shows that no other cardinality can occur. Taking $\pi = g = \iota$ gives rank $r_0 = 1$ and cardinality p . If π is any non-affine permutation and $g = \pi$, then $[\iota]$ and $[\pi]$ are linearly independent modulo the constants, giving rank $r_0 = 2$ and cardinality p^2 . Such a permutation exists for $p \geq 5$, for example the transposition in Lemma 5.2: an affine map fixing three points must be the identity, whereas this transposition is not. Finally, the pair in Lemma 5.2 has rank $r_0 = 3$ and gives cardinality p^3 . $\square$

Example 5.5. For $p = 5$, with $\pi = (0, 1, 2, 4, 3)$ and $g = (1, 2, 0, 3, 4)$, the cube

$$|c_{r,j,k}\rangle = \frac{1}{\sqrt{5}} (\omega_5^{rs+j\pi(s)+kg(s)})_{s \in \mathbb{Z}_5}, \quad r, j, k \in \mathbb{Z}_5,$$

is a QLC(5) with cardinality 125. All inner products are fifth-root geometric sums and the verification is exact; a machine check (residuals $\sim 10^{-16}$, no projective coincidences among the 125 entries) is provided in the accompanying scripts, together with the analogous verifications for $p = 7, 11, 13, 17$.

Remark 5.6. The tri-clock cube is layered in the sense of Section 3: $C = C(V_g, Q)$ where $V_g = \sum_s \omega^{g(s)} |s\rangle\langle s|$ and Q is the layer $k = 0$,

$$|q_{r,j}\rangle = \frac{1}{\sqrt{p}} (\omega^{rs+j\pi(s)})_{s \in \mathbb{Z}_p}.$$

All entries of Q are flat with respect to the computational basis, in accordance with Theorem 3.4. Note also that all p^3 entries of C are flat: the cube is, as a whole, unbiased to one fixed basis.

Corollary 5.7 (cf. [8, Thm. 3.1]). *For every prime $p \geq 5$ the square Q of Remark 5.6 (with π any non-affine permutation of $\mathbb{Z}_p$, e.g. the transposition of Lemma 5.2) is a QLS(p) of maximal cardinality p^2 all of whose entries are unbiased to the computational basis.*

Proof. Rows and columns are orthonormal as in Theorem 5.1. Two entries coincide iff $(r - r')\iota + (j - j')\pi \in \mathbb{F}_p \mathbf{1}$; if $j \neq j'$ this exhibits π as an affine function $\pi(s) = (j - j')^{-1}(c - (r - r')s)$, and if $j = j'$, $r \neq r'$ it makes $(r - r')\iota$ constant; both are impossible. Non-affine permutations exist for $p \geq 5$ because $p! > p(p - 1)$. $\square$

Remark 5.8. Up to relabelling, the square Q is the family of maximal-cardinality quantum Latin squares constructed, for *all* orders $v \geq 4$, by Zang, Zheng, Tian and Shan [8, Thm. 3.1], who phase the columns of the Fourier matrix by the columns of a row-permuted Fourier matrix. The present formulation adds two points. First, at prime orders the non-affineness of the permutation π is *necessary* as well as sufficient for maximal cardinality (the proof of Corollary 5.7

is an equivalence). Second, all entries of these squares are unbiased to the computational basis, and by Theorem 3.4 this flatness is exactly the property required of the layer of a quantum Latin cube. This identifies the specific structural reason why this family, rather than any other known family of maximal-cardinality squares, lifts to the cubes of Corollary 5.3. By Musto's correspondence between quantum Latin squares and mutually unbiased bases [3], flat squares are also a natural source of MUB-type structures; we do not pursue this here.

We can now prove the main theorem.

Proof of Theorem 1.3. For $v = 1$ the unique QLC(1) has cardinality $1 = v^3$. For $v \in \{2, 3\}$ maximal cardinality is impossible by Theorem 6.2 below. For $v \geq 4$ composite, apply Corollary 4.5; for $v \geq 5$ prime, apply Corollary 5.3. $\square$

6. SMALL ORDERS, THE ORDER-FOUR OBSTRUCTION, AND OPEN PROBLEMS

Lemma 6.1. *A 3×3 unitary matrix with zero diagonal is monomial: it has exactly one nonzero entry in every row and every column.*

Proof. Let the columns be $b_0 = (0, \beta, \gamma)^\top$, $b_1 = (\delta, 0, \varepsilon)^\top$, $b_2 = (\zeta, \eta, 0)^\top$. Orthogonality of the three pairs of columns gives $\bar{\gamma}\varepsilon = \bar{\beta}\eta = \bar{\delta}\zeta = 0$. If $\gamma = 0$, then $|\beta| = 1$, so $\eta = 0$, so $|\zeta| = 1$, so $\delta = 0$, and the matrix is monomial. If instead $\varepsilon = 0$, then $|\delta| = 1$, so $\zeta = 0$, so $|\eta| = 1$, so $\beta = 0$, and again the matrix is monomial. $\square$

Theorem 6.2. *Every QLC(2) has cardinality 2, and every QLC(3) has cardinality 3: quantum Latin cubes of orders 2 and 3 are classical up to a global unitary. In particular the cardinality spectra of quantum Latin cubes of orders 2 and 3 are $\{2\}$ and $\{3\}$, and maximal cardinality v^3 is unattainable.*

Proof. *Order 2.* Each layer $\{|c_{i,j,k}\rangle\}_{i,j}$ (fixed k) is a QLS(2), so by Lemma 2.1 it carries the two vectors of an orthonormal basis of $\mathbb{C}^2$. The file condition forces $|c_{i,j,1}\rangle \perp |c_{i,j,0}\rangle$, and in $\mathbb{C}^2$ the orthogonal complement of a unit vector is projectively unique; hence layer 1 carries the same two projective vectors as layer 0, and $c(C) = 2$.

Order 3. By Lemma 2.1, layer k is classical: there are a Latin square L_k of order 3 and an orthonormal basis

$$B_k = \{|b_0^{(k)}\rangle, |b_1^{(k)}\rangle, |b_2^{(k)}\rangle\}$$

with $|c_{i,j,k}\rangle \propto |b_{L_k(i,j)}^{(k)}\rangle$. After a global unitary we may assume $B_0 = \{|e_0\rangle, |e_1\rangle, |e_2\rangle\}$ is the computational basis. We claim $B_1 = B_0$ projectively.

For $y \in \{0, 1, 2\}$ let $N(y) = \{L_0(i, j) : L_1(i, j) = y\}$. The file through a cell (i, j) with $L_1(i, j) = y$ contains $|c_{i,j,0}\rangle$, so $|b_y^{(1)}\rangle \perp |e_x\rangle$ for every $x \in N(y)$.

Suppose first that $|N(y_0)| = 1$ for some y_0 , say $N(y_0) = \{x_0\}$. The three cells of the symbol y_0 in L_1 then carry all three occurrences of x_0 in L_0 , so $x_0 \notin N(y)$ for $y \neq y_0$ and $N(y_1), N(y_2) \subseteq \{x_1, x_2\}$. If some $|N(y_1)| = 2$, then each of x_1, x_2 occurs at least once among the cells of y_1 and hence at most twice among those of y_2 , while y_2 has three cells; consequently $N(y_2) = \{x_1, x_2\}$ as well. But then $|b_{y_1}^{(1)}\rangle$ and $|b_{y_2}^{(1)}\rangle$ are both orthogonal to $|e_{x_1}\rangle$ and $|e_{x_2}\rangle$, hence both proportional to $|e_{x_0}\rangle$, contradicting their orthogonality. Therefore in this case $|N(y)| = 1$ for all y , and $y \mapsto N(y)$ is a bijection (two symbols of L_1 cannot pair exclusively with the same symbol of L_0 , which occurs only three times). The matrix of B_1 in computational coordinates then has, after relabelling the columns, zero diagonal, so it is monomial by Lemma 6.1: $B_1 = B_0$ projectively.

Otherwise $|N(y)| \geq 2$ for every y , and each $|b_y^{(1)}\rangle$, being orthogonal to two distinct computational vectors, is proportional to the third: again $B_1 \subseteq B_0$, hence $B_1 = B_0$ as both are bases.

The same argument gives $B_2 = B_0$, so all 27 entries are computational up to phase and $c(C) = 3$. $\square$

Theorem 6.2 is consistent with the numerical exploration of the QLC(3) variety: 60 out of 60 random alternating-projection runs converged, all of them to classical solutions of cardinality 3 (script `verify_qlc3_rigidity.py`).

Proposition 6.3 (Order four obstruction). *There is no pair of permutations π, g of $\mathbb{Z}_4$ such that $s \mapsto \alpha s + \beta \pi(s) + \gamma g(s) \pmod 4$ is non-constant for every $(\alpha, \beta, \gamma) \in \mathbb{Z}_4^3 \setminus \{(0, 0, 0)\}$. Consequently the tri-clock recipe of Theorem 5.1 cannot produce a cube of maximal cardinality at order 4.*

Proof. Let $\rho : \mathbb{Z}_4 \rightarrow \mathbb{F}_2$ be reduction modulo 2. For any permutation h of $\mathbb{Z}_4$ the vector $\rho \circ h \in \mathbb{F}_2^{\mathbb{Z}_4} \cong \mathbb{F}_2^4$ is *balanced* (two 0s and two 1s), since h hits two even and two odd values. In the quotient space $\mathbb{F}_2^4 / \langle \mathbf{1} \rangle$ (dimension 3) the six balanced vectors fall into exactly three classes, namely $[0101]$, $[0011]$ and $[0110]$, and these satisfy $[0011] + [0110] = [0101]$; together with 0 they form a 2-dimensional subspace. The classes $[\rho \circ \iota] = [0101]$, $[\rho \circ \pi]$ and $[\rho \circ g]$ therefore lie in a common 2-dimensional space, so they are linearly dependent over $\mathbb{F}_2$: there are $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \mathbb{F}_2$, not all zero, and $\varepsilon_0 \in \mathbb{F}_2$ with $\varepsilon_1 \rho(s) + \varepsilon_2 \rho(\pi(s)) + \varepsilon_3 \rho(g(s)) + \varepsilon_0 = 0$ in $\mathbb{F}_2$ for all s . Multiplying by 2 and using $2\rho(x) \equiv 2x \pmod 4$, we obtain

$$2\varepsilon_1 s + 2\varepsilon_2 \pi(s) + 2\varepsilon_3 g(s) \equiv -2\varepsilon_0 \pmod 4 \quad \text{for all } s,$$

a constant combination with $(\alpha, \beta, \gamma) = (2\varepsilon_1, 2\varepsilon_2, 2\varepsilon_3) \neq (0, 0, 0)$. $\square$

Remark 6.4. For composite v the tri-clock cube of Theorem 5.1 is still a QLC(v) for any permutations π, g of $\mathbb{Z}_v$ (the line computations only use bijectivity). Its cardinality is the size of the image of the $\mathbb{Z}_v$ -module homomorphism

$$\mathbb{Z}_v^3 \longrightarrow \mathbb{Z}_v^{\mathbb{Z}_v} / \langle \mathbf{1} \rangle, \quad (\alpha, \beta, \gamma) \longmapsto [\alpha \iota + \beta \pi + \gamma g].$$

In particular its cardinality divides v^3 . It has maximal cardinality if and only if $\alpha \iota + \beta \pi + \gamma g$ is non-constant modulo v for all $(\alpha, \beta, \gamma) \neq (0, 0, 0)$ — equivalently, if for every prime $q \mid v$ the reductions $\{\mathbf{1}, \iota, \pi, g\} \pmod q$ are linearly independent over $\mathbb{F}_q$ as functions on $\mathbb{Z}_v$. Proposition 6.3 shows this fails for $v = 4$; an exhaustive computer search confirms that no pair of permutations of $\mathbb{Z}_4$ works. For $v = 6$ pairs do exist; for example, $\pi = (0, 1, 2, 3, 5, 4)$, $g = (0, 1, 2, 4, 3, 5)$ give a tri-clock QLC(6) of cardinality 216, as verified by the accompanying script. Thus the twisted-product construction of Section 4 and the tri-clock construction overlap at least at order 6. A characterisation of the orders v admitting tri-clock pairs would be of independent combinatorial interest.

Problem 6.5. *Determine the possible image sizes of the module homomorphism in Remark 6.4 as π, g range over permutations of $\mathbb{Z}_v$. In particular, characterise the composite orders v for which there are permutations π, g such that $\alpha \iota + \beta \pi + \gamma g$ is non-constant modulo v for every nonzero $(\alpha, \beta, \gamma) \in \mathbb{Z}_v^3$. Equivalently, by the Chinese remainder theorem, characterise when the reductions of $\{\mathbf{1}, \iota, \pi, g\}$ remain linearly independent over every prime field $\mathbb{F}_q$ with $q \mid v$. The obstruction above rules out $v = 4$, while the example in Remark 6.4 shows that $v = 6$ is already possible.*

6.1. The certified low spectrum of QLS(5). Write $\text{Spec}_{\text{QLS}}(n)$ for the set of cardinalities attained by quantum Latin squares of order n . The order-five case already displays substantial gaps. The following result combines the exact and computer-assisted conclusions of the order-five investigation. Here *certified* means that, after the first-row normalisation, the remaining possibilities are reduced to a finite list of support-edge types and every type is eliminated by an explicitly recorded obstruction from Lemma 6.6. No floating-point feasibility test enters the nonexistence proof.

Lemma 6.6 (Support–orthogonality obstructions). *Let $x_1, \dots, x_t$ be distinct non-computational projective vectors of $\mathbb{C}^5$, with prescribed coordinate supports of size at least two and with some prescribed orthogonalities. Each of the following configurations is impossible.*

- (a) *q pairwise orthogonal vectors have their supports contained in a coordinate subspace of dimension less than q .*
- (b) *Distinct x, y, z lie in a common 2-space and satisfy $x \perp y \perp z$.*
- (c) *Orthogonal x, y span a coordinate 2-space V , while $z \perp x, y$ has fewer than two support coordinates outside V .*
- (d) *A vector x with 2-element support is orthogonal to y , and their supports meet in exactly one coordinate, without y vanishing at that coordinate.*
- (e) *Distinct u, w have the same 3-element support S and two common orthogonal neighbours a, b whose 2-element supports lie in S , meet in one coordinate, and together cover S .*

Proof. Part (a) is the dimension bound for pairwise orthogonal nonzero vectors. For (b), the orthogonal complement of y inside the common 2-space is a line containing both x and z . In (c), the projection of z onto V is orthogonal to the basis $\{x, y\}$, so z is supported outside V and therefore degenerates to a computational vector. For (d), if $\text{supp}(x) = \{i, j\}$ and the intersection is $\{i\}$, then $0 = \langle x, y \rangle = \overline{x_i} y_i$; both coordinates of the non-computational vector x are nonzero, hence $y_i = 0$. Finally, in (e) the support conditions make a, b linearly independent, and the orthogonal complement of their span in $\mathbb{C}^S$ is a line containing both u and w . $\square$

Theorem 6.7 (Computer-assisted certified low spectrum of order five).

$$\text{Spec}_{\text{QLS}}(5) \cap [5, 12] = \{5, 7, 12\}.$$

More precisely, cardinalities 5, 7 and 12 are attained, whereas 6, 8, 9, 10 and 11 are not.

Proof. Cardinality 5 is attained by every classical Latin square. For cardinality 7, take the Latin square with rows

$$01234, \quad 10342, \quad 23401, \quad 34120, \quad 42013.$$

Its first two rows and first two columns contain an intercalate on symbols 0, 1; rotating those two symbols by a generic angle in their coordinate 2-space introduces exactly two new projective vectors. Construction 6.9 gives cardinality 12. The exclusion of $6 = 5 + 1$ is the general $n + 1$ nonexistence lemma of [9, Lem. 3.1].

It remains to exclude 8, 9, 10, 11. Normalize the first row to the computational basis. A square of cardinality $c = 5 + t$ then has t new projective classes on the remaining 20 cells. Its *active graph* is the bipartite graph on the four lower rows and five columns whose edges are the cells carrying new classes. Every complete labelling determines coordinate supports for the new classes and a forced orthogonality graph. The deterministic certificate has three layers: enumerate active graphs up to row and column permutations; enumerate all admissible label patterns with new labels introduced in canonical first-occurrence order; and classify the resulting support–edge types by Lemma 6.6. The exhaustive terminal statistics, counted after symmetry reduction at the active-graph layer, are

c	t	graph types	label patterns	support–edge types
8	3	5	34 560	directly classified
9	4	14	1 587	97
10	5	40	9 010	1 724
11	6	220	201 297	46 808

and no type survives. Thus none of 8, 9, 10, 11 is attainable. The table is not a numerical approximation: each row records a finite certificate whose leaves are disposed of by Lemma 6.6. $\square$

Remark 6.8 (Proof boundary). The exact constructions and Lemma 6.6 are ordinary mathematical proofs. The exclusions of cardinalities 8–11 additionally depend on the exhaustiveness and correctness of the enumeration programs. The machine-readable outputs record zero unclassified terminal types; the public repository cited in the computational note contains the entry-point scripts, certificates, and a reproducibility guide.

Construction 6.9 (An exact cardinality-12 square). *Let $e_0, \dots, e_4$ be the computational vectors and define the following unnormalised integer vectors:*

$$\begin{aligned} a &= (0, -1, 0, 2, 1), & b &= (0, 1, 0, 1, -1), & c &= (0, 1, 0, 0, 1), \\ d &= (0, -1, 0, 2, -5), & h &= (0, 2, 0, 1, 0), & f &= (0, 1, 0, -2, 0), \\ g &= (0, 1, 0, 0, -1). \end{aligned}$$

After normalising these seven vectors, the array

$$Q_{12} = \begin{pmatrix} e_0 & e_1 & e_2 & e_3 & e_4 \\ a & e_0 & b & c & e_2 \\ d & e_2 & a & e_0 & h \\ h & e_4 & e_0 & e_2 & f \\ e_2 & e_3 & c & g & e_0 \end{pmatrix}$$

is a QLS(5) of cardinality exactly 12. All row and column orthogonality checks reduce to integer dot products before normalisation.

Remark 6.10. The next unresolved low value is 13. A tangent-space calculation at Q_{12} gives projective tangent nullity 4 after fixing the first row, and the nullity remains 4 under every tested single-cell split of a repeated class; this is evidence of first-order rigidity, not a nonexistence proof. High-precision numerical candidates of cardinalities 21 and 24 have also been found, while extensive stratified searches suggest the conjectural full spectrum $\text{Spec}_{\text{QLS}}(5) = \{5, 7, 12, 21, 24, 25\}$. At present we make no exact claim about 13, $\dots$, 20, 22, 23.

Problem 6.11. *Determine the cardinality spectrum of quantum Latin cubes: for which $c \in [v, v^3]$ does a QLC(v) of cardinality c exist? Theorem 6.2 settles the question for $v \leq 3$: the spectra are $\{2\}$ and $\{3\}$. For $v \geq 4$ both endpoints are attained (the classical cube, and Theorem 1.3), and Lemma 3.5 shows that every flat QLS(v) stratum of cardinality c' with N twist-orbit classes contributes the value vN to the spectrum. Exploratory alternating projections on the QLC(5) variety (script `qlc_experiments.py`) reached, besides the value 125 of Corollary 5.3, the cardinalities 21, 25 and 45. These are numerical candidates rather than theorems, but they indicate that the spectrum of QLC(5) is nontrivial in its middle range. The candidate 21 would necessarily be non-layered by Corollary 3.6; and Theorem 5.1 shows that neither 21 nor 45 can arise from the tri-clock construction, whose order-five cardinalities are restricted to 5, 25 and 125. The corresponding spectrum problem for squares is an active subject [5, 9, 10]; Theorem 6.7 determines its order-five low-cardinality interval, but the middle range remains open. Even the first attainable cardinality strictly above v , and whether long intervals of cardinalities occur for any fixed v , are unknown in general.*

Problem 6.12. *Corollary 5.7 produces, in every prime dimension $p \geq 5$, quantum Latin squares of maximal cardinality all of whose p^2 entries are unbiased to a fixed basis. Which cardinalities are achievable by such flat quantum Latin squares? By Theorem 3.4 this question, together with the orbit quotient in Lemma 3.5, governs all layered quantum Latin cubes. In 173 converged random least-squares searches over the flat QLS(5) variety we observed only the two cardinalities 5 and 25, suggesting that the flat spectrum may be much sparser than the unrestricted one.*

Problem 6.13. *Complete the order-five square spectrum. The first exact target is the value 13, where the same support–edge certificate framework applies but the number of dense active graphs becomes the bottleneck. A second target is to replace the numerical candidates of cardinalities 21 and 24 by exact constructions, or to prove that they cannot be rationalised inside any low-dimensional algebraic family related to Q_{12} .*

Problem 6.14. *Determine the complete spectra of $QLS(5)$ and $QLS(6)$. For order five the certified beginning is Theorem 6.7. For order six, the general $n + 1$ obstruction excludes 7, the dense theorem of [10] applies only from order 12 onward, and the explicit order-six constructions of [11] show that the small spectrum has new isolated values beyond the classical and maximal endpoints. A complete classification of these two orders would test whether the sparse behaviour of $QLS(5)$ is exceptional or the first instance of a broader small-order phenomenon.*

Concluding perspective. Theorem 1.3 closes the existence problem at the maximal endpoint: apart from the rigid small orders 2 and 3, every order admits a quantum Latin cube with all entries projectively distinct. The proof also isolates the structural requirement behind the layered method. A square can serve as a layer precisely when it is flat in an eigenbasis whose associated unitary has cyclic simple spectrum; maximality then becomes an orbit-separation problem. This separates two tasks that were previously intertwined and provides a framework for future spectrum constructions. Corollary 3.6 and Theorem 5.1 also turn this framework into an obstruction theory: layered cardinalities are multiples of v , and the prime-order tri-clock spectrum is exactly $\{p, p^2, p^3\}$. On the constructive side, Corollary 4.4 shows that the composite-order endpoint is not isolated: every nontrivial factorisation $v = mn \geq 6$ supplies a factorial family of admissible twists, with at least $v\varphi(v)(v-4)!$ choices of eigenvalue assignment.

The result does not determine non-maximal cube cardinalities, nor does it classify maximal cubes up to equivalence. Those problems require control of orbit collisions rather than their complete avoidance. The certified order-five square calculation illustrates one possible complementary tool: finite support–orthogonality obstructions can detect gaps that explicit constructions alone do not reveal.

Computational note. All constructions used in the proofs are exact. The tri-clock entries are roots of unity, the composite construction uses Fourier entries together with the transcendental phase $e^{i\sqrt{2}r}$ only to separate projective classes, and the cardinality-12 square uses normalised integer vectors. The corresponding checks reduce to finite geometric sums, phase-comparison identities, or integer dot products.

The computer-assisted nonexistence result for $\text{Spec}_{QLS}(5) \cap [8, 11]$ is finite and deterministic: the scripts enumerate active graphs, label patterns and support–edge types, then attach one of the obstructions of Lemma 6.6 to each terminal type. The numerical least-squares and alternating-projection experiments mentioned in Remarks 6.10 and Problems 6.11, 6.12 are exploratory only.

The verification scripts, selected machine-readable certificates, compiled manuscript, and a reproducibility guide are available at

<https://github.com/okmath2026/Quantum-Latin-Cubes>.

The guide identifies the scripts used for the cube constructions, the order-three and order-four checks, and the order-five square-spectrum certificates. The corresponding certificate outputs record zero unclassified terminal types, and the companion report in the repository records the full certificate statistics.

REFERENCES

- [1] D. Goyeneche, Z. Raissi, S. Di Martino and K. Życzkowski, *Entanglement and quantum combinatorial designs*, Phys. Rev. A **97** (2018), 062326.
- [2] B. Musto and J. Vicary, *Quantum Latin squares and unitary error bases*, Quantum Inf. Comput. **16** (2016), 1318–1332.
- [3] B. Musto, *Constructing mutually unbiased bases from quantum Latin squares*, Electron. Proc. Theor. Comput. Sci. **236** (2017), 108–126.
- [4] I. Nechita and J. Pillet, *SudoQ — a quantum variant of the popular game*, Quantum Inf. Comput. **21** (2021), 781–799.
- [5] J. Paczos, M. Wierzbiński, G. Rajchel-Mieldzioć, A. Burchardt and K. Życzkowski, *Genuinely quantum solutions of the game Sudoku and their cardinality*, Phys. Rev. A **104** (2021), 042423.
- [6] Y. Zhang and H. Cao, *Quantum Latin squares with maximal cardinality*, Discrete Math. **349** (2026), 114863.
- [7] Y. Zhang, Y. Zhang, M. Lv and H. Cao, *The maximal cardinality of quantum Latin squares and quantum Latin cubes*, J. Combin. Des. **34** (2026), 169–183.
- [8] Y. Zang, M. Zheng, Z. Tian and X. Shan, *On the cardinalities of quantum Latin squares*, arXiv:2508.01972 (2025).
- [9] Y. Zhang, X. Wang and L. Ji, *Quantum Latin squares with all possible cardinalities*, J. Combin. Des. **34** (2026), 378–387.
- [10] Y. Zhang and L. Ji, *Quantum Latin squares of order $6m$ with all possible cardinalities*, arXiv:2601.09132 (2026).
- [11] Z. Xu, *Three quantum Latin squares of order 6 with cardinalities 13, 15 and 17*, arXiv:2605.15540 (2026).

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